Background modeling

General information:

- Follow the instructions in the examples one by one and in the case of the tasks set out, implement them (detailed task is always presented at the end of the instructions), answer the questions and include it in the report. Put the report into the folder on eKursy.
- Extract RAR packed file from course page to your working folder

Example – Estimation of the background

- Run the `example3.py`
- Note the implementation of the loop that generates `frameIds` vector elements.
- Note the method for determining the mean and median in the image vector `numpy.mean` and `numpy.median` object functions.
- Note the difference between the mean and median results, discuss the complexity and computation time of the both algorithm.

```
import numpy as np
import cv2

# Open Video

cap = cv2.VideoCapture('highway.mp4')

number_of_frame=0
frameIds = []



while number_of_frame < 101:
    frameIds.append(number_of_frame)
    number_of_frame=number_of_frame+1


# Store selected frames in an array
frames = []

for fid in frameIds:
    cap.set(cv2.CAP_PROP_POS_FRAMES, fid)
    ret, frame = cap.read()
    frames.append(frame)

# Calculate the mean along the time axis
meanFrame = np.mean(frames, axis=0).astype(dtype=np.uint8)

# Calculate the median along the time axis
medianFrame = np.median(frames, axis=0).astype(dtype=np.uint8)



# Display median frame
cv2.imshow('median frame', medianFrame)
cv2.imwrite('medianFrame.jpg', medianFrame, [cv2.IMWRITE_JPEG_QUALITY, 100])

# Display mean frame
cv2.imshow('mean frame', meanFrame)
cv2.imwrite('meanFrame.jpg', meanFrame, [cv2.IMWRITE_JPEG_QUALITY, 100])
```

```
cv2.waitKey(0)

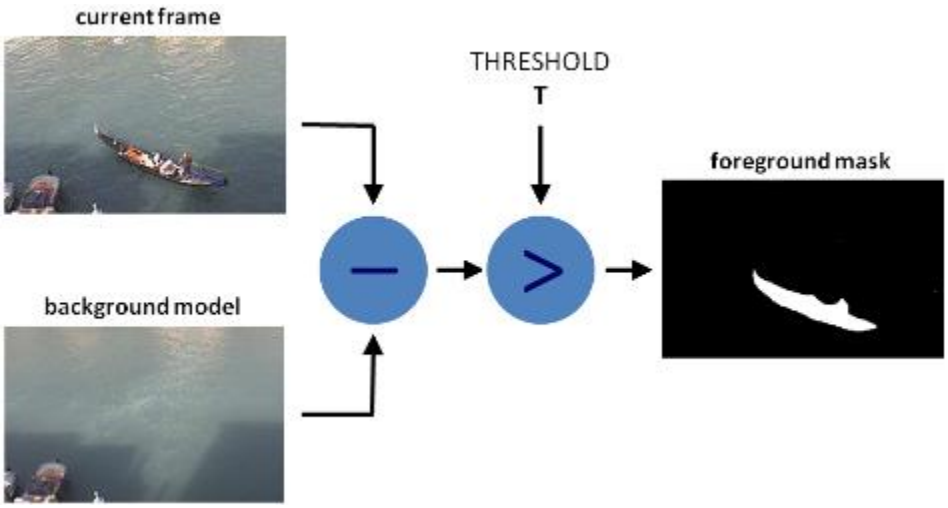
# When everything is done, release the video capture object

cap.release()
# Closes all frames
cv2.destroyAllWindows()
```

Example – Foreground mask generation

Background subtraction (BS) is a common and widely used technique for generating a foreground mask (namely, a binary image containing the pixels belonging to moving objects in the scene) by using static cameras.

As the name suggests, BS calculates the foreground mask performing a subtraction between the current frame and a background model, containing the static part of the scene or, more in general, everything that can be considered as background given the characteristics of the observed scene.



Background modeling consists of two main steps:

1. Background Initialization;
2. Background Update.

In the first step, an initial model of the background is computed, while in the second step that model is updated in order to adapt to possible changes in the scene.

- Use the `medianFrame.jpg` from the output of the previous example.
- Run the `example4.py` file downloaded from course page
- Note the ***`absdiff`*** and ***`threshold`*** functions.

```
import cv2
import numpy as np


currentFrame = cv2.imread('currentFrame.jpg',cv2.IMREAD_UNCHANGED)
backgroundModel = cv2.imread('medianFrame.jpg',cv2.IMREAD_UNCHANGED)


first_gray = cv2.cvtColor(currentFrame, cv2.COLOR_BGR2GRAY)
gray_frame = cv2.cvtColor(backgroundModel, cv2.COLOR_BGR2GRAY)

difference = cv2.absdiff(first_gray, gray_frame)
_, difference = cv2.threshold(difference, 25, 255, cv2.THRESH_BINARY)

cv2.imshow("Current frame", currentFrame)
cv2.imshow("Background Model", backgroundModel)
cv2.imshow("thresholded difference", difference)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Example – Background subtractor – Mixture of Gaussians, KNN

- Run the `example5.py` file downloaded from eKursy
- Note the definition of subtractor: `createBackgroundSubtractorMOG2()` and `createBackgroundSubtractorKNN`

    - Inside the parenthesis we can change the value of the subtractor. History is the number of the last frame that are taken into consideration (by default 120). The threshold value is the value used when computing the difference to extract the background. A lower threshold will find more differences with the advantage of a more noisy image. Detectshadows is a function of the algorithm that can remove the shadows if enabled.There are no right or wrong values, you need to try different settings to see what best fits your need.
    - `subtractor = cv2.createBackgroundSubtractorMOG2(history=120, varThreshold=50, detectShadows=True)`

- *https://opencv-python-tutroals.readthedocs.io/en/latest/py_tutorials/py_video/py_bg_subtraction/py_bg_subtraction.html*

```
import cv2 as cv


# MOG2
backSub = cv.createBackgroundSubtractorMOG2()

# KNN
#backSub = cv.createBackgroundSubtractorKNN()


cap = cv.VideoCapture('highway.mp4')
if not cap.isOpened:
    print('Unable to open file')
    exit(0)

while True:
    ret, frame = cap.read()

    if ret is True:


        fgMask = backSub.apply(frame)


        cv.imshow('Frame', frame)
        cv.imshow('FG Mask', fgMask)

        if cv.waitKey(25) & 0xFF == ord('q'):
            break


    else:
        break

capture.release()
# Closes all frames
cv.destroyAllWindows()
```

Task

- Prepare the source code of the program – background subtractor, in which the goal is to extract the mask of foreground elements, as is done in example 5. The average background should be used as the model of the estimated background instead the MOG and KNN (the option should also be the median filtration as an estimation of the background).

- Reporting on the implementation of the task:
    - program code in the .py file to run